

ADIB ROUF

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EDUCATION

UNIVERSITY OF CONNECTICUT, COLLEGE OF ENGINEERING

August 2024 - May 2027

Storrs Mansfield, CT

B.S. in Computer Science

Relevant Coursework: Computer Architecture, Algorithms & Complexities, Data Structures, Object-Oriented Design, Discrete Systems, Systems Programming, Artificial Intelligence, Machine Learning

Concentration: Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: Python, SQL, C, JavaScript, HTML/CSS

Frameworks & Tools: Snowflake, Supabase, Flask, React, GitHub, AWS, Azure

EXPERIENCE

DATA ENGINEERING INTERN

May 2026 – Present

Travelers Insurance

Hartford, CT

- Rebuilt a legacy VB Script tool in Python and SQL, resolving 200,000+ user-role records across 8,500+ groups to make the workflow maintainable for the team
- Built a parameterized Python script to pull cross-domain group membership across production, EU, and admin environments
- Classified access by NTIC ID and person-vs-machine type to support internal access reviews

L'SPACE NPWEE PARTICIPANT

January 2026 – April 2026

NASA - National Aeronautics and Space Administration | [Certificate](#)

Remote, Based in AZ

- Evaluated performance criteria for a proposed spaceflight materials testing protocol against mission-duration constraints, structuring findings into testable requirements
- Authored technical assumptions and requirement specifications adopted into NASA's final proposal submission

UNDERGRAD RESEARCH INTERN

March 2025 – December 2025

University of Connecticut | [XAI4AV](#)

Storrs Mansfield, CT

- Built a full-stack ML pipeline for uploading video and saliency data, triggering HPC-based LLM inference, and returning model decision heatmaps
- Designed an async status pipeline (pending → processing → complete) connecting the frontend, Supabase, and HPC compute backend
- Shipped upload, queuing, and results UI end-to-end, returning AI-generated and original video side-by-side

PROJECTS

ML SERVING PLATFORM

June 2026 – Present

- Built an ML inference platform with FastAPI and an async job queue, decoupling request submission from model execution via polling-based status tracking
- Implemented dynamic request batching in the worker pipeline, improving inference throughput by 88% (56 → 106 req/sec) under concurrent load

DATA AGGREGATION AND INSIGHTS PLATFORM

December 2024 – March 2025

- Processed 28,900+ XML documents by building a Python pipeline that extracted structured data into pandas DataFrames.
- Loaded cleaned data into Snowflake by creating an automated workflow for storage, querying, and analysis
- Improved query performance by 30% by restructuring data processing logic and optimizing how records were prepared for Snowflake